

claude-windows / dca1ff63-3aec-42ab-897f-41970003f4e1

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-khelsutra\dca1ff63-3aec-42ab-897f-41970003f4e1\subagents\workflows\wf_f40bdb2b-ed3\agent-a37c36c070a47a389.jsonl`  
- SHA-256: `897f5971d829ac0331492527963db1e11077cd4da98b90764b9e10006e860ebd`  
- Source modified: `2026-06-21T18:50:56+00:00`  
- Imported at: `2026-07-05T16:48:21+00:00`  
- project: `wf_f40bdb2b-ed3`  
- session_id: `dca1ff63-3aec-42ab-897f-41970003f4e1`
```

Transcript

1. user (2026-06-21T18:49:28.141Z)

You are a meticulous senior reviewer. Project: KhelSutra. A3 makes the dependency-free marketing site (site/) localize its recording-education + marketing copy to Kannada/Hindi via a vanilla t()/data-i18n shim that reads shared catalogs at repo-root /locales (mirrored to site/public/locales). The shim (site/public/i18n.js) added a data-i18n.html form that sets innerHTML from catalog values (these values are authored by the project ? never user input ? and contain only tags). Pages bundle Noto via Google Fonts. en text stays in the HTML as the progressive-enhancement default; the shim overrides for kn/hi.

Files to read (absolute):

```
shim: C:/Users/avidu/Projects/khelsutra-fu-wt/site/public/i18n.js  
capture.html: C:/Users/avidu/Projects/khelsutra-fu-wt/site/public/capture.html  
index.html: C:/Users/avidu/Projects/khelsutra-fu-wt/site/public/index.html  
shim test: C:/Users/avidu/Projects/khelsutra-fu-wt/site/src/i18n.shim.test.js  
assets test: C:/Users/avidu/Projects/khelsutra-fu-wt/site/src/assets.test.js  
en catalog: C:/Users/avidu/Projects/khelsutra-fu-wt/locales/en/common.json  
kn catalog: C:/Users/avidu/Projects/khelsutra-fu-wt/locales/kn/common.json
```

Review ONLY through this lens:

TEST COVERAGE: read both test files. What A3 behavior is NOT covered that should be? (e.g. data-i18n.html fallback-on-missing-key, font-stack assertion robustness, kn render of a rich key end-to-end, drift between HTML default and en catalog for the new keys). Suggest concrete missing test cases. Do not propose tests that already exist.

Read the relevant files, then report concrete findings (file + precise issue + a concrete fix). Be adversarial but precise ? only real issues, no style nitpicks unless they cause bugs. If the code is clean for this lens, return an empty findings array.

2. assistant (2026-06-21T18:49:32.593Z)

I'll read all the relevant files to review test coverage for A3.

3. user (2026-06-21T18:49:33.029Z)

```
1      // Unit tests for the marketing-site i18n shim (public/i18n.js, tracker B3). Proves the  
shim mirrors  
2      // the SPA's react-i18next contract: same negotiation order, ICU-lite  
interpolation/plurals, en  
3      // fallback, and a data-i18n walker that NEVER leaves a raw key. Tested against the REAL  
shared  
4      // catalogs (/locales) ? so this also proves both surfaces consume one source of truth.  
5      import { test } from "node:test";  
6      import assert from "node:assert/strict";  
7      import { readFileSync } from "node:fs";  
8      import { join } from "node:path";  
9      import { JSDOM } from "jsdom";  
10     import { negotiate, interpolate, translate, applyDom, SUPPORTED, DEFAULT } from
```

```

"../public/i18n.js";
11
12     const LOCALES = join(import.meta.dirname, "..", "..", "locales");
13     const load = (loc) => JSON.parse(readFileSync(join(LOCALES, loc, "common.json"),
"utf8"));
14     const catalogs = { en: load("en"), kn: load("kn"), hi: load("hi") };
15
16     test("negotiation order matches the SPA: persisted -> ?lang -> navigator -> en", () => {
17         assert.equal(negotiate({ stored: "kn", query: "hi", nav: ["en-US"] }), "kn",
"persisted pref wins");
18         assert.equal(negotiate({ query: "hi", nav: ["en-US"] }), "hi", "?lang beats
navigator");
19         assert.equal(negotiate({ nav: ["kn-IN", "en-IN"] }), "kn", "navigator picks the first
supported");
20         assert.equal(negotiate({}), "en", "nothing -> en");
21     });
22
23     test("region subtags normalize and unsupported degrades to en (never a locale we can't
serve)", () => {
24         assert.equal(negotiate({ nav: ["hi-IN"] }), "hi", "hi-IN -> hi");
25         assert.equal(negotiate({ nav: ["ta-IN"] }), "en", "Tamil (unsupported) -> en");
26         assert.equal(negotiate({ stored: "es" }), "en", "es is SPA-only ? the trilingual site
degrades it to en");
27         assert.deepEqual([...SUPPORTED], ["en", "kn", "hi"]);
28         assert.equal(DEFAULT, "en");
29     });
30
31     test("ICU-lite interpolation: {name} substitution", () => {
32         assert.equal(interpolate("over {n}s", { n: 10 }), "over 10s");
33         assert.equal(interpolate("Couldn't load: {error}", { error: "boom" }), "Couldn't load:
boom");
34     });
35
36     test("ICU-lite plurals via Intl.PluralRules (en/kn 2-form), # = the number", () => {
37         const en = catalogs.en.selection.count; // "{count, plural, one {# rally selected}
other {# rallies selected}}"
38         assert.equal(interpolate(en, { count: 1 }, "en"), "1 rally selected");
39         assert.equal(interpolate(en, { count: 5 }, "en"), "5 rallies selected");
40         // Kannada is also one/other ? the shim picks the arm from the catalog, no hard-coded
rules.
41         assert.equal(translate(catalogs, "kn", "selection.count", { count: 1 }),
catalogs.kn.selection.count.match(/one \{([^\}]*\)\}/)[1].replace("#", "1"));
42     });
43
44     test("translate: active locale, then en fallback, then undefined (caller keeps the DOM
default)", () => {
45         assert.equal(translate(catalogs, "kn", "app.title"), catalogs.kn.app.title); // real
Kannada
46         assert.equal(translate(catalogs, "en", "lang.kn"), "?????"); // shared catalog autonym
47         // A3: site.* marketing is now translated for kn (no longer the en seed).
48         assert.equal(translate(catalogs, "kn", "site.capture.title"),
catalogs.kn.site.capture.title);
49         assert.notEqual(catalogs.kn.site.capture.title, catalogs.en.site.capture.title, "kn
capture.title must be translated, not an en seed");
50         // en fallback when a key is absent from the active locale (synthetic ? real catalogs
have full parity).
51         assert.equal(translate({ en: { only: { en: "fallback" } }, kn: {} }, "kn", "only.en"),
"fallback");
52         assert.equal(translate(catalogs, "kn", "does.not.exist"), undefined, "missing
everywhere -> undefined");
53     });
54
55     test("applyDom walks data-i18n, sets attrs, marks the switcher, and never shows a raw
key", () => {
56         const { document } = new JSDOM(`<!doctype html><body>

```

```

57     <span id="auto" data-i18n="lang.kn">PLACEHOLDER</span>
58     <span id="keep" data-i18n="does.not.exist">KEEP ME</span>
59     <input id="ph" data-i18n="site.capture.title" data-i18n-attr="placeholder" />
60     <button id="b-en" data-lang="en">English</button>
61     <button id="b-kn" data-lang="kn">????</button>
62 </body>`).window;
63
64     applyDom(document, catalogs, "kn");
65
66     assert.equal(document.getElementById("auto").textContent, "????", "data-i18n sets
textContent from the shared catalog");
67     assert.equal(document.getElementById("keep").textContent, "KEEP ME", "a missing key
leaves the HTML default ? never a raw key");
68     assert.equal(document.getElementById("ph").getAttribute("placeholder"),
catalogs.kn.site.capture.title, "data-i18n-attr targets an attribute (kn value for the kn
locale)");
69     assert.equal(document.getElementById("b-kn").getAttribute("aria-pressed"), "true",
"active locale button is pressed");
70     assert.equal(document.getElementById("b-en").getAttribute("aria-pressed"), "false",
"inactive locale button is not pressed");
71     assert.ok(document.getElementById("b-kn").classList.contains("is-active"), "active
button gets a non-color selection class");
72     });
73
74     test("data-i18n-html sets innerHTML and preserves inline <b> markup (A3 recording-
education)", () => {
75         const { document } = new JSDOM(`<!doctype html><body>
76             <p id="rich" data-i18n-html="site.record.place.body">Behind <b>one baseline</b>
?</p>
77             <li id="kept" data-i18n-html="does.not.exist">KEEP <b>this</b></li>
78         </body>`).window;
79
80         // en: innerHTML equals the catalog value verbatim (with its <b> tags), not escaped
text.
81         applyDom(document, catalogs, "en");
82         assert.equal(document.getElementById("rich").innerHTML,
catalogs.en.site.record.place.body);
83         assert.ok(document.getElementById("rich").querySelector("b"), "the <b> must be a real
element, not escaped text");
84         assert.equal(document.getElementById("kept").innerHTML, "KEEP <b>this</b>", "a missing
html key leaves the HTML default");
85
86         // kn: swaps to the Kannada value, still carrying its <b> tags.
87         applyDom(document, catalogs, "kn");
88         assert.equal(document.getElementById("rich").innerHTML,
catalogs.kn.site.record.place.body);
89         assert.match(document.getElementById("rich").innerHTML, /<b>.*<\b>/s, "kn rich value
keeps <b> markup");
90     });
91
92     test("A3: the kn/hi catalogs carry REAL recording-education translations (not English
seeds)", () => {
93         // Kannada script range U+0C80?U+0CFF; Devanagari U+0900?U+097F. A representative
spread of the
94         // recording keys must be in-script (proves A3 actually translated, not B3's en seed).
95         const KN = /[?-\?]/, HI = /[?-\?]/;
96         for (const key of ["record.intro", "record.place.title", "capture.step1.b3",
"capture.consent", "nav.requestAccess"]) {
97             const path = key.split(".");
98             const kn = path.reduce((o, k) => o[k], catalogs.kn.site);
99             const hi = path.reduce((o, k) => o[k], catalogs.hi.site);
100             assert.match(kn, KN, `kn site.${key} should be in Kannada script (a leftover English
seed would not match)`);
101             assert.match(hi, HI, `hi site.${key} should be in Devanagari script`);
102         }

```

```

103      // A brand/token-free key must be FULLY in-script (no stray English) ? strongest anti-
seed check.
104      assert.doesNotMatch(catalogs.kn.site.capture.consent, /[A-Za-z]/, "kn capture.consent
has no business containing Latin letters");
105      assert.doesNotMatch(catalogs.hi.site.capture.consent, /[A-Za-z]/, "hi capture.consent
has no business containing Latin letters");
106      // Brand + literal tokens are intentionally preserved across languages.
107      assert.match(catalogs.kn.site.record.intro, /KheISutra/, "brand stays in Latin");
108      assert.match(catalogs.hi.site.record.settings.body, /1080p\+|30 fps/, "technical
tokens stay literal");
109    });
110

```

4. user (2026-06-21T18:49:33.978Z)

```

1      // Static-site integrity ? the landing page + capture checklist + OG image ARE the
public
2      // product. server.test.js proves the VM host *serves* "/", "/capture", "/og.png"; this
3      // guards the CONTENT contract so a broken edit (deleted asset, broken form wiring, dead
4      // capture link, missing/!absolute OG image) fails CI instead of shipping to the live
site.
5      import { test } from "node:test";
6      import assert from "node:assert/strict";
7      import { readFileSync, statSync } from "node:fs";
8      import { join } from "node:path";
9
10     const PUBLIC = join(import.meta.dirname, "..", "public");
11     const read = (f) => readFileSync(join(PUBLIC, f), "utf8");
12     const size = (f) => statSync(join(PUBLIC, f)).size;
13     const enCatalog = JSON.parse(readFileSync(join(import.meta.dirname, "..", "..",
"locales", "en", "common.json"), "utf8"));
14
15     test("core public assets exist and are non-empty", () => {
16       for (const f of ["index.html", "capture.html", "og.png"]) {
17         assert.ok(size(f) > 0, `public/${f} is missing or empty`);
18       }
19     });
20
21     test("landing page wires the Request-Access form to the API", () => {
22       const html = read("index.html");
23       assert.match(html, /id="af"/, "the #af form is gone");
24       assert.match(html, /\api\/request-access/, "form no longer posts to /api/request-
access");
25       assert.match(html, /name="company"[^>]*tabindex="-1"/, "honeypot field removed (bots
will get through)");
26       for (const field of ["segment", "has_gpu", "storage"]) {
27         assert.match(html, new RegExp(`name="${field}"`), `form lost the '${field}' field (a
demand signal)`);
28       }
29     });
30
31     test("submitting the form shows an obvious in-flight state (no silent freeze)", () => {
32       const html = read("index.html");
33       assert.match(html, /Sending?/, "submit should swap to a 'Sending?' state while the
request is in flight");
34       assert.match(html, /aria-busy/, "submit should set aria-busy for assistive tech during
the request");
35       assert.match(html, /class="spin"/, "the in-flight spinner is gone");
36       assert.match(html, /\spin\{/, "the spinner style is gone");
37       // the button must be RE-ENABLED on failure so the user isn't stuck on a frozen
'Sending?'
38       assert.match(html, /btn\.disabled\s*=\s*false/, "submit button isn't restored on the
failure path");
39     });
40

```

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41     test("OG + canonical meta point at the live https origin", () => {
42         const html = read("index.html");
43         assert.match(html,
/property="og:image"\s+content="https:\\\\khelsutra\\.guru\\/og\\.png"/, "og:image must be the
absolute live URL");
44         assert.match(html, /property="og:url"\s+content="https:\\\\khelsutra\\.guru"/, "og:url
drift");
45     });
46
47     test("the sample reel is GATED behind sign-up + starts at a RANDOM point (not shown
openly)", () => {
48         const html = read("index.html");
49         assert.match(html, /ajTCZ7pJW_o/, "the sample-reel video id is gone");
50         assert.match(html, /youtube-nocookie\\.com/, "reel should use the privacy nocookie
host");
51         assert.match(html, /id="reel-reveal"/, "the on-signup reveal host is gone");
52         assert.match(html, /see it in action/i, "the 'see it in action' sign-up teaser is
gone");
53         assert.match(html, /reel-count/, "the auto-close countdown is gone");
54         assert.match(html, /REEL_SECS = 30/, "the reel preview should be ~30s");
55         assert.match(html, /getDuration\\(\\)/, "random start should derive from the real video
duration");
56         assert.match(html, /Math\\.random\\(\\)/, "the start point should be randomized");
57         assert.match(html, /seekTo\\(/, "the player should seek to the random start");
58         // Must NOT be an openly-rendered public section ? it only appears after a successful
submit.
59         assert.doesNotMatch(html, /<section id="reel"/, "the reel must be sign-up-gated, not
an open public section");
60         assert.doesNotMatch(html, /id="reel-play"/, "the open click-to-play facade should be
removed");
61     });
62
63     test("the TOOL screencast is self-hosted, footage-free, and revealed only after sign-
up", () => {
64         const html = read("index.html");
65         assert.match(html, /\\/video\\/tool-cuj\\.mp4/, "the tool-operation screencast reference
is gone");
66         assert.match(html, /function revealAfterSignup/, "the post-signup reveal function is
gone");
67         assert.match(html, /revealAfterSignup\\(\\)/, "the reveal must be wired to a successful
submit");
68         // GATED: the only reference to the tool screencast lives inside the reveal function
(injected on
69         // signup), not as an open autoplay element rendered for every visitor.
70         assert.ok(
71             html.indexOf("/video/tool-cuj.mp4") > html.indexOf("function revealAfterSignup"),
72             "the tool screencast must be inside the post-signup reveal, not shown openly",
73         );
74         // the self-hosted clip exists and is non-trivial (a real recording, not an empty
placeholder)
75         assert.ok(size("video/tool-cuj.mp4") > 50_000, "tool-cuj.mp4 is missing or too small
to be the real screencast");
76     });
77
78     test("the post-signup reveal keeps its watchability affordances (manual close + play-
gated auto-close)", () => {
79         // Content contract only ? the BEHAVIOUR (close works, auto-close holds while the reel
plays) is
80         // proven against the real page in ux.dom.test.js, so we don't pin brittle
implementation strings.
81         const html = read("index.html");
82         assert.match(html, /id="reel-close"/, "the manual ? close button is gone");
83         assert.match(html, /onStateChange/, "the reel's play-state must still gate the auto-
close");
84     });

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```

85
86     test("the capture checklist is linked and is a real page", () => {
87         assert.match(read("index.html"), /href="\capture(\.html)?"/, "footer capture-
capture checklist link is gone");
88         const cap = read("capture.html");
89         assert.match(cap, /<html/i, "capture.html is not HTML");
90         assert.match(cap, /capture checklist/i, "capture.html lost its checklist content");
91     });
92
93     test("the academy flyers (en/kn/hi) exist, are language-correct, and are linked from the
homepage", () => {
94         const html = read("index.html");
95         // All three are linked from the homepage footer (the owner ask: the flyers must be
findable).
96         assert.match(html, /href="\flyer"/, "homepage doesn't link the English academy
flyer");
97         assert.match(html, /href="\flyer-kn"/, "homepage doesn't link the Kannada academy
flyer");
98         assert.match(html, /href="\flyer-hi"/, "homepage doesn't link the Hindi academy
flyer");
99
100        // The three flyer pages exist, are HTML, and carry the right language + script.
101        const en = read("flyer.html");
102        assert.match(en, /<html lang="en"/i, "flyer.html should be lang=en");
103        const kn = read("flyer-kn.html");
104        assert.match(kn, /<html lang="kn"/i, "flyer-kn.html should be lang=kn");
105        assert.match(kn, /[?-\?]/, "flyer-kn.html should contain Kannada text");
106        const hi = read("flyer-hi.html");
107        assert.match(hi, /<html lang="hi"/i, "flyer-hi.html should be lang=hi");
108        assert.match(hi, /[?-\?]/, "flyer-hi.html should contain Devanagari (Hindi) text");
109    });
110
111    test("each flyer shows a screenshot of the tool rendered in that language", () => {
112        for (const [flyer, img] of [["flyer.html", "studio-en.png"], ["flyer-kn.html",
"studio-kn.png"], ["flyer-hi.html", "studio-hi.png"]]) {
113            assert.ok(size(`img/${img}`) > 0, `public/img/${img} (the localized UI shot) is
missing or empty`);
114            assert.match(read(flyer), new RegExp(`img/${img}`), `${flyer} should embed its
localized UI screenshot /img/${img}`);
115        }
116    });
117
118    test("the academy flyers are also linked from the top nav and the own-model page", () =>
{
119        const html = read("index.html");
120        // Desktop top nav carries a Flyer entry (alongside Own your model).
121        const nav = html.match(/<nav class="nav-links"[\s\S]*?<\nav>/);
122        assert.ok(nav, "the primary nav block is gone");
123        assert.match(nav[0], /href="\flyer"/, "the top nav doesn't link the Flyer");
124        // own-model surfaces all three languages.
125        const om = read("own-model.html");
126        assert.match(om, /href="\flyer"/, "own-model doesn't link the English flyer");
127        assert.match(om, /href="\flyer-kn"/, "own-model doesn't link the Kannada flyer");
128        assert.match(om, /href="\flyer-hi"/, "own-model doesn't link the Hindi flyer");
129    });
130
131    test("the landing page surfaces how to record reliably (so users can set up with
confidence)", () => {
132        const html = read("index.html");
133        assert.match(html, /id="record"/, "the 'record it right' section is gone");
134        assert.match(html, /one match, one court/i, "the #1 single-court capture rule should
be on the homepage");
135        // the gist must carry the load-bearing capture cues + a prominent link to the full
checklist
136        for (const cue of [/1080p/, /30 fps/i, /[Aa]udio ON/, /baseline/, /[Nn]o panning/]) {

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137     assert.match(html, cue, `the recording gist is missing a key cue: ${cue}`);
138   }
139   assert.match(html, /href="\capture"[^>]*>Full 1-page capture checklist/i, "a
prominent link to the full checklist is gone");
140   });
141
142   test("the 'How do I join?' FAQ points at the request-access form, not email", () => {
143     const html = read("index.html");
144     const m = html.match(/How do I join?<\summary>(<p>[\s\S]*?<\p>)/);
145     assert.ok(m, "the 'How do I join?' FAQ is gone");
146     assert.match(m[1], /request access/i, "the join FAQ should send people to the request-
access form");
147     assert.match(m[1], /href="#join"/, "the join FAQ should link to the #join form");
148     assert.doesNotMatch(m[1], /mailto/, "the join FAQ should NOT tell people to email (use
the form)");
149   });
150
151   test("the own-model page exists, is honestly scoped, and is linked from the landing
page", () => {
152     const om = read("own-model.html");
153     assert.ok(om.length > 0, "own-model.html is missing or empty");
154     for (const id of ["annotate", "train", "setup"]) {
155       assert.match(om, new RegExp(`id="${id}"`), `own-model is missing the #${id}
section`);
156     }
157     assert.match(om, /github\.com\/Khelsutra\/rally-annotator/, "the GitHub link to the
annotator is gone");
158     assert.match(om, /Read this honestly/i, "the honest framing fence is gone");
159     assert.match(om, /open pipeline|in build/i, "training must be honestly scoped as in-
build, not one-click");
160     assert.match(om, /name="robots" content="noindex"/, "own-model should be noindex");
161     assert.match(read("index.html"), /href="\own-model"/, "the landing page should link
to /own-model");
162     assert.match(om, /<a href="\/">? KhelSutra home<\a>/, "own-model needs an obvious
back-to-home link (not just browser-back)");
163
164     // a ready-made default works without training (keeps non-tech users engaged); BYO-
training is optional
165     assert.match(om, /ready-made default|out of the box|don't have to train/i, "must
reassure a default works without training");
166     assert.match(om, /optional/i, "training your own should be framed as optional");
167
168     // own-model shows the annotation TOOL screenshots (no people) ? real captures, files
present
169     for (const img of ["annotate-enable.png", "annotate-tool.png"]) {
170       assert.match(om, new RegExp(`/img/${img}`), `own-model should embed /img/${img}`);
171       assert.ok(size(`/img/${img}`) > 0, `public/img/${img} is missing or empty`);
172     }
173     assert.match(om, /<img[>]+alt="/i, "screenshots must have alt text");
174
175     // the sample REEL stays gated behind request-access ? NOT embedded/playable here, no
footage shown
176     assert.doesNotMatch(om, /<iframe/i, "the reel must NOT be embedded here ? it's gated
behind request-access");
177     assert.doesNotMatch(om, /youtube/i, "no YouTube embed on own-model; the reel is shown
only after request-access");
178     assert.doesNotMatch(om, /i\.ytimg\.com/, "don't show the video thumbnail (players)
openly");
179     assert.match(om, /sample[\s\S]{0,80}request access|request access[\s\S]{0,80}sample/i,
"point users to request access for a sample output");
180   });
181
182   // ---- B3: the trilingual marketing surface (shared t()/data-i18n shim + persistent
switcher) -----
183   const I18N_PAGES = ["index.html", "capture.html", "own-model.html"];

```

```

184
185     test("every localizable page loads the shared i18n shim", () => {
186         for (const f of I18N_PAGES) {
187             assert.match(read(f), /<script[^>]+type="module"[^>]+src="\/i18n\.js"/, `${f}
doesn't load the i18n shim (/i18n.js)`);
188         }
189     });
190
191     test("every localizable page has a persistent en/kn/hi language switcher", () => {
192         for (const f of I18N_PAGES) {
193             const html = read(f);
194             assert.match(html, /class="langsw"[^>]*role="group"[^>]*aria-label="Language"/,
`${f} is missing the language switcher group`);
195             for (const lang of ["en", "kn", "hi"]) {
196                 assert.match(html, new RegExp(`<button[^>]+data-lang="${lang}"`), `${f} switcher
lacks the ${lang} button`);
197             }
198             // The switcher labels are driven by the SHARED catalog (lang.*) ? proves cross-
surface consumption.
199             assert.match(html, /data-
lang="kn"[^>]*data-i18n="lang\.kn"|data-i18n="lang\.kn"[^>]*data-lang="kn"/, `${f} switcher
isn't wired to the shared catalog (lang.kn)`);
200             // The Kannada/Hindi autonyms must be present in their own scripts (system-font
fallback renders them).
201             assert.match(html, /?????/, `${f} switcher lacks the Kannada autonym`);
202             assert.match(html, /?????/, `${f} switcher lacks the Hindi autonym`);
203         }
204     });
205
206     test("marketing content is wired for translation via data-i18n, and every key exists in
the en catalog", () => {
207         const dig = (obj, key) => key.split(".").reduce((o, k) => (o == null ? o : o[k]),
obj);
208         const used = new Set();
209         for (const f of I18N_PAGES) {
210             const html = read(f);
211             const keys = [...html.matchAll(/data-i18n="([^\"]+)"/g)].map((m) => m[1]);
212             assert.ok(keys.length >= 3, `${f} should wire at least a few strings via data-i18n
(found ${keys.length})`);
213             keys.forEach((k) => used.add(k));
214         }
215         // Every data-i18n key across the site must resolve in the en source catalog (no
dangling keys -> the
216         // shim would otherwise leave the English HTML default, but a typo'd key is a silent
bug).
217         for (const key of used) {
218             assert.notEqual(dig(enCatalog, key), undefined, `data-i18n="${key}" has no entry in
the en catalog`);
219         }
220         // The shared site.* marketing namespace is present (the B3 deliverable ? en authored,
kn/hi later).
221         assert.ok(enCatalog.site && enCatalog.site.nav && enCatalog.site.hero, "the en catalog
is missing the site.* marketing namespace");
222     });
223
224     test("the en catalog value matches the HTML default for representative data-i18n keys
(no drift)", () => {
225         // The HTML carries the English default text (SEO/no-JS); the en catalog must mirror
it so an en
226         // visitor renders identically and a future kn/hi translation simply overrides the
same string.
227         assert.match(read("capture.html"), /<h1 data-i18n="site\.capture\.title">1-page
capture checklist<\h1>/);
228         assert.equal(enCatalog.site.capture.title, "1-page capture checklist");
229         assert.match(read("index.html"), /data-i18n="site\.nav\.requestAccess">Request

```



```

access</>);
230     assert.equal(enCatalog.site.nav.requestAccess, "Request access");
231   });
232
233   // ---- A3: recording-education localized + Noto fonts bundled for kn/hi rendering
-----
234   test("A3: every marketing page bundles Noto Kannada + Devanagari so kn/hi render (not
tofu)", () => {
235     for (const f of ["index.html", "capture.html", "own-model.html"]) {
236       const html = read(f);
237       assert.match(html, /Noto\+Sans\+Kannada/, `${f} doesn't load Noto Sans Kannada`);
238       assert.match(html, /Noto\+Sans\+Devanagari/, `${f} doesn't load Noto Sans
Devanagari`);
239       assert.match(html, /--font:[^;]*'Noto Sans Kannada'[^;]*'Noto Sans Devanagari'/,
`${f} font stack must fall back to Noto for Indic glyphs`);
240     }
241   });
242
243   test("A3: the recording-education content is wired for translation (homepage gist +
capture page)", () => {
244     const dig = (obj, key) => key.split(".").reduce((o, k) => (o == null ? o : o[k]),
obj);
245     // Both the rich (data-i18n-html, inline <b>) and plain (data-i18n) forms are used,
and EVERY key
246     // across BOTH forms resolves in the en source catalog (a typo'd key would silently
show English).
247     for (const f of ["index.html", "capture.html"]) {
248       const html = read(f);
249       const keys = [...html.matchAll(/data-i18n(?:-html)?="([^\"]+)"/g)].map((m) =>
m[1]).filter((k) => k.startsWith("site."));
250       for (const key of keys) {
251         assert.notEqual(dig(enCatalog, key), undefined, `${f}: data-i18n key "${key}" has
no en catalog entry`);
252       }
253     }
254     // The capture page localizes the load-bearing recording cues via the rich (html)
form.
255     const cap = read("capture.html");
256     assert.match(cap, /data-i18n-html="site.capture.step1.b1"/, "capture step-1
placement bullet isn't localizable");
257     assert.match(cap, /data-i18n-html="site.capture.step2.b3"/, "capture audio bullet
isn't localizable");
258     // The homepage gist localizes its cards.
259     assert.match(read("index.html"), /data-i18n-html="site.record.settings.body"/,
"homepage settings card isn't localizable");
260     // en catalog's recording values keep the inline <b> markup the HTML relies on.
261     assert.match(enCatalog.site.record.place.body, /<b>.*<\b>/s, "en record.place.body
lost its <b> markup");
262     assert.match(enCatalog.site.capture.step2.b1, /<b>1080p or higher<\b>/, "en
capture.step2.b1 markup drift");
263   });
264

```

5. user (2026-06-21T18:49:34.123Z)

```

1     // KhelSutra marketing-site i18n shim (tracker B3) ? vanilla, dependency-free, zero
build.
2     //
3     // Mirrors the SPA's react-i18next contract (web/src/i18n/index.ts) for the static site:
4     //   - the SAME shared catalogs (repo-root /locales, copied into /public/locales ?
D-CATALOG-HOME);
5     //   - the SAME negotiation order: persisted (localStorage `kheIsutra.lang`) -> ?lang=
-> navigator -> en;
6     //   - ICU-lite interpolation `{name}` + plurals `{n, plural, one {?} other {?}}` via
Intl.PluralRules

```

```

7      //      (CLDR-correct for en/kn/hi ? the 2-form site languages ? and any locale);
8      //      - a missing key degrades to en, then leaves the HTML's English default ? NEVER a
raw key.
9      //
10     // The pure helpers are exported for the unit test (src/i18n.shim.test.js); boot() runs
only in a browser.
11
12     export const SUPPORTED = ["en", "kn", "hi"]; // the trilingual marketing site (es/da/id
are SPA-only)
13     export const STORAGE_KEY = "khelsutra.lang"; // shared with the SPA, so a choice carries
across surfaces
14     export const DEFAULT = "en";
15
16     const base = (code) => (code ? String(code).toLowerCase().split("-")[0] : ""); // "kn-
IN" -> "kn"
17     const supported = (code) => (SUPPORTED.includes(base(code)) ? base(code) : null);
18
19     // Negotiation ? EXACTLY the SPA order. Anything unsupported is skipped; nothing matches
-> en.
20     export function negotiate({ stored, query, nav } = {}) {
21         const navList = Array.isArray(nav) ? nav : nav ? [nav] : [];
22         for (const candidate of [stored, query, ...navList]) {
23             const ok = supported(candidate);
24             if (ok) return ok;
25         }
26         return DEFAULT;
27     }
28
29     const dig = (obj, key) => key.split(".").reduce((o, k) => (o && typeof o === "object" ?
o[k] : undefined), obj);
30
31     // Resolve one plural block "{var, plural, one {?} other {?}}" (balanced braces, # = the
number).
32     // Returns the string unchanged when there's no parseable plural block (so interpolate()
terminates).
33     function plural(msg, vars, locale) {
34         const m = msg.match(/\{(\w+),\s*plural,/);
35         if (!m) return msg;
36         let depth = 0, end = -1;
37         for (let i = m.index; i < msg.length; i++) {
38             if (msg[i] === "{") depth++;
39             else if (msg[i] === "}" && --depth === 0) { end = i; break; }
40         }
41         if (end < 0) return msg;
42         const inner = msg.slice(m.index + m[0].length, end);
43         const n = Number(vars[m[1]] ?? 0);
44         const cat = new Intl.PluralRules(locale).select(n);
45         const arms = {};
46         for (const a of inner.matchAll(/(\w+)\s*\{/g)) {
47             let d = 1, j = a.index + a[0].length, e = -1;
48             for (; j < inner.length; j++) { if (inner[j] === "{") d++; else if (inner[j] === "}"
&& --d === 0) { e = j; break; } }
49             if (e < 0) break;
50             arms[a[1]] = inner.slice(a.index + a[0].length, e);
51         }
52         const chosen = (arms[cat] ?? arms.other ?? "").replace(/#/g, String(n));
53         return msg.slice(0, m.index) + chosen + msg.slice(end + 1);
54     }
55
56     // ICU-lite: resolve every plural block, then simple {name} interpolation.
57     export function interpolate(msg, vars = {}, locale = DEFAULT) {
58         if (typeof msg !== "string") return msg;
59         let out = msg;
60         for (let i = 0; i < 20; i++) { const next = plural(out, vars, locale); if (next ===
out) break; out = next; }

```

```

61     return out.replace(/\{(\w+)\}/g, (_, k) => (k in vars ? String(vars[k]) : `${k}`));
62 }
63
64 // active locale -> en fallback. undefined when absent in both (caller keeps the DOM's
English default).
65 export function translate(catalogs, locale, key, vars) {
66     let msg = dig(catalogs[locale], key);
67     if (msg == null) msg = dig(catalogs[DEFAULT], key);
68     return msg == null ? undefined : interpolate(msg, vars, locale);
69 }
70
71 // Walk i18n-tagged nodes under `root`. Three forms (missing key -> skip, leaving the
HTML default):
72 //   data-i18n="key"                                -> textContent
73 //   data-i18n="key" data-i18n-attr="x"              -> setAttribute(x, ?) (placeholder / aria-
label / alt ?)
74 //   data-i18n-html="key"                            -> innerHTML (for catalog-authored inline
markup like <b>;
75 //                                                    values are our own content, never user
input)
76 export function applyDom(root, catalogs, locale) {
77     for (const el of root.querySelectorAll("[data-i18n], [data-i18n-html]")) {
78         const htmlKey = el.getAttribute("data-i18n-html");
79         const val = translate(catalogs, locale, htmlKey || el.getAttribute("data-i18n"));
80         if (val == null) continue;
81         const attr = el.getAttribute("data-i18n-attr");
82         if (htmlKey) el.innerHTML = val;
83         else if (attr) el.setAttribute(attr, val);
84         else el.textContent = val;
85     }
86     for (const el of root.querySelectorAll("[data-lang]")) {
87         const on = el.getAttribute("data-lang") === locale;
88         el.setAttribute("aria-pressed", String(on));
89         el.classList.toggle("is-active", on);
90     }
91 }
92
93 // ---- browser glue (skipped under the plain-node unit test, which has no document)
----
94 const readCookie = (name) => {
95     const m = (document.cookie || "").match(new RegExp("(?:^|; )" + name.replace(/\./g,
"\\.") + "=[^;]*"));
96     return m ? decodeURIComponent(m[1]) : null;
97 };
98
99 function persist(locale) {
100     try { localStorage.setItem(STORAGE_KEY, locale); } catch { /* private mode */ }
101     // Mirror to a cookie so server.js (VM host) can honor the persisted pref server-side
(no FOUc).
102     try { document.cookie = `${STORAGE_KEY}=${locale};path=/;max-
age=31536000;samesite=lax`; } catch { /* ignore */ }
103 }
104
105 async function loadCatalog(loc) {
106     try { const r = await fetch(`/locales/${loc}/common.json`); return r.ok ? await
r.json() : {}; } catch { return {}; }
107 }
108
109 async function render(locale) {
110     const catalogs = { en: await loadCatalog("en") };
111     if (locale !== "en") catalogs[locale] = await loadCatalog(locale);
112     document.documentElement.lang = locale;
113     applyDom(document, catalogs, locale);
114 }
115

```

```

116     async function boot() {
117         let stored = null;
118         try { stored = localStorage.getItem(STORAGE_KEY); } catch { /* ignore */ }
119         if (!stored) stored = readCookie(STORAGE_KEY); // server-set or cross-store persisted
pref
120         const query = new URLSearchParams(location.search).get("lang");
121         const nav = navigator.languages && navigator.languages.length ?
[...navigator.languages] : [navigator.language];
122         const locale = negotiate({ stored, query, nav });
123         await render(locale);
124         document.addEventListener("click", (e) => {
125             const btn = e.target.closest && e.target.closest("[data-lang]");
126             if (!btn) return;
127             e.preventDefault();
128             const loc = supported(btn.getAttribute("data-lang")) || DEFAULT;
129             persist(loc);
130             render(loc);
131         });
132     }
133
134     if (typeof document !== "undefined" && typeof window !== "undefined") {
135         if (document.readyState === "loading") document.addEventListener("DOMContentLoaded",
boot);
136         else boot();
137     }
138

```

6. assistant (2026-06-21T18:49:43.098Z)

Now let me read the catalogs and the relevant HTML to check for drift and coverage gaps.

7. user (2026-06-21T18:49:44.061Z)

```

1      {
2          "app": {
3              "badge": "pick your rallies · alpha",
4              "title": "Pick the rallies for your reel",
5              "intro": "Filter by <b>rally length</b> ? e.g. <i>?top 15 longest?</i> ? then fine-
tune by tapping cards. Everything starts <b>selected</b>; just drop the ones you don't want."
6          },
7          "lang": {
8              "label": "Language",
9              "en": "English",
10             "kn": "?????",
11             "hi": "?????",
12             "es": "Español",
13             "da": "Dansk",
14             "id": "Bahasa Indonesia"
15         },
16         "bulk": {
17             "selectAll": "Select all",
18             "clear": "Clear",
19             "invert": "Invert"
20         },
21         "mode": {
22             "demo": "These are sample rallies ? process a video above to work on a real match.",
23             "loading": "Finding the rallies in your video?",
24             "error": "Couldn't load this video's rallies: {error}",
25             "loaded": "Found <b>{count}</b> rallies in your video."
26         },
27         "query": {
28             "placeholder": "e.g. over 10s · top 15 longest · shortest first",
29             "ariaLabel": "Filter rallies by length, count or order",
30             "apply": "Apply",
31             "notAvailable": "not available yet",

```

```

32     "notAvailableTitle": "?{token}? needs {kind} detection ? not available yet",
33     "notAvailableHelp": "Shot type, players and score aren't available yet ? filter by
rally length instead.",
34     "unreadable": "Couldn't read that ? try ?over 10s? or ?top 15 longest?.",
35     "hint": "Type over 10s for rallies longer than 10 seconds ? or tap a sample below.",
36     "try": "Sample queries:",
37     "support": "Filtering by <b>rally length</b> is fully supported. Shot-count is
<x>experimental</x> (under development).",
38 },
39     "grid": {
40         "empty": "No rallies to show yet.",
41         "rally": "Rally {n}",
42         "inReel": "Rally {n}, in reel",
43         "notInReel": "Rally {n}, not in reel",
44         "start": "start",
45         "long": "long",
46         "shots": "~{count} shots · experimental",
47         "shotsTitle": "Shot count is experimental (under development) and may be
inaccurate{conf}. Filter by rally length instead.",
48         "shotsConfidence": " ? confidence: {conf}"
49     },
50     "footer": {
51         "count": "{count, plural, one {# rally} other {# rallies}}",
52         "summaryRest": "selected · {total} total",
53         "reelName": "Reel name",
54         "reelNamePlaceholder": "my-highlights",
55         "build": "Build reel ({count})",
56         "building": "Building?",
57         "floorOne": "<b>Your only selected rally</b> has fewer than {min} shots ? it would
be skipped, so the build would fail. Pick a longer rally.",
58         "floorAll": "<b>All {count} selected rallies</b> have fewer than {min} shots ?
they'd all be skipped, so the build would fail. Pick longer rallies.",
59         "floorSomeLead": "{count, plural, one {# rally} other {# rallies}}",
60         "floorSomeRest": "with fewer than {min} shots will be skipped when building (they're
too short).",
61     },
62     "result": {
63         "ready": "Your reel is ready",
64         "failed": "Build failed",
65         "building": "Stitching your reel?",
66         "buildingReassure": "Joining your selected rallies into one video ? this usually
only takes a moment.",
67         "dismiss": "Dismiss",
68         "download": "Download {filename}"
69     },
70     "build": {
71         "noVideo": "These are sample rallies ? process a real video first to build a reel."
72     },
73     "ingest": {
74         "title": "Process a video",
75         "intro": "Point KhelSutra at a video and it finds the rallies for you ? this can
take a few minutes.",
76         "uriPlaceholder": "gs://your-bucket/match.mp4",
77         "uriAriaLabel": "Video location ? a gs:// bucket URL or a path on the engine host",
78         "hint": "Paste a cloud-storage link (gs://?), or a path to a video on the machine
running KhelSutra.",
79         "submit": "Process video",
80         "submitting": "Submitting?",
81         "cancel": "Cancel",
82         "working": "Processing your video?",
83         "progress": "{progress}",
84         "retry": "Try again",
85         "reassure": "Safe to wait ? KhelSutra keeps working even if your connection is slow;
a long video can take several minutes.",
86         "statusFailed": "Couldn't process",

```

```

87     "chooseFile": "Choose a video file",
88     "uploadAria": "Choose a video file to upload",
89     "fileHint": "Pick an .mp4 or .mov from this device ? it uploads, then KhelSutra
finds the rallies.",
90     "orLink": "Or use a cloud link",
91     "uploadingStatus": "Uploading?",
92     "uploadProgress": "Upload progress"
93 },
94     "errors": {
95         "networkError": "Can't reach KhelSutra. Is it running?",
96         "invalidUri": "Enter a video location first ? a gs:// link or a path on the machine
running KhelSutra.",
97         "emptyResult": "Processing finished but found no rallies in this video. Try a
clearer match recording.",
98         "timeout": "This is taking longer than expected. The job may still be running ?
reload with the video link, or try again.",
99         "engineGeneric": "Something went wrong. Please try again."
100     },
101     "tap": {
102         "title": "Tap to filter:",
103         "longest": "Longest first",
104         "shortest": "Shortest first",
105         "over10": "Over 10 seconds",
106         "over20": "Over 20 seconds",
107         "top5": "Top 5 longest",
108         "top10": "Top 10 longest"
109     },
110     "site": {
111         "nav": {
112             "how": "How it works",
113             "why": "Why it's hard",
114             "ownModel": "Own your model",
115             "flyer": "Flyer",
116             "roadmap": "Roadmap",
117             "faq": "FAQ",
118             "requestAccess": "Request access"
119         },
120         "hero": {
121             "eyebrow": "Invite-only alpha · Bengaluru",
122             "lead": "KhelSutra finds every rally in your session and strips out the knock-ups,
walks and gaps ? so a full match becomes a clean highlight reel you can watch and download. No
editing, no scrubbing.",
123             "ctaPrimary": "Request alpha access",
124             "ctaHow": "See how it works"
125         },
126         "record": {
127             "kicker": "Record it right",
128             "title": "Five minutes of setup ? reliably great reels",
129             "intro": "Highlight quality starts with the recording. Set the camera up once like
this and KhelSutra has everything it needs to find the rallies.",
130             "goldenRuleLead": "The golden rule: one match, one court, in full frame ? and
don't move the camera.",
131             "goldenRuleRest": "A single court in frame gives the best results today; busy
multi-court footage is the hard case we're still sharpening.",
132             "place": {
133                 "title": "Place the camera",
134                 "body": "Behind <b>one baseline</b>, elevated (a high tripod or a balcony),
tilted down ~30° ? the <b>whole court</b> in frame, players never cropped."
135             },
136             "settings": {
137                 "title": "Settings",
138                 "body": "<b>1080p+</b>, landscape, <b>30 fps</b> for badminton (60 fps for table
tennis). <b>Audio ON</b> ? the shuttle's ?thwack? helps find rallies."
139             },
140             "still": {

```

```

141         "title": "Keep it still",
142         "body": "Fixed tripod or mount. <b>No panning, no zooming</b> during play ? a
steady, full-court view is what the tool is tuned for."
143     },
144     "upload": {
145         "title": "Record & upload",
146         "body": "Even lighting, then record the <b>whole session</b> and upload it as-is
? KhelSutra finds the rallies and does the cutting."
147     },
148     "checklistCta": "Full 1-page capture checklist ?"
149 },
150 "capture": {
151     "title": "1-page capture checklist",
152     "intro": "Great highlights start with a steady, full-court recording. Five minutes
of setup makes a big difference to your reels.",
153     "ruleLead": "The golden rule: one match, one court, in full frame ? and don't move
the camera.",
154     "ruleRest": "Multi-court footage is the hard case we're still sharpening; a single
court in frame gives the best results today.",
155     "step1": {
156         "title": "Place the camera",
157         "b1": "Behind <b>one baseline</b>, elevated ? a high tripod, a few metres up, or
a balcony.",
158         "b2": "Tilt down about <b>30°</b>; keep the <b>whole court</b> in frame with a
little margin and net headroom.",
159         "b3": "Players should never leave the frame."
160     },
161     "step2": {
162         "title": "Settings",
163         "b1": "<b>1080p or higher</b>, shot in <b>landscape</b>.",
164         "b2": "<b>30 fps</b> for badminton/pickleball; <b>60 fps</b> for table tennis.",
165         "b3": "<b>Audio ON</b> (don't mute) ? keep the mic unobstructed, wind/noise
reduction off."
166     },
167     "step3": {
168         "title": "Keep it still",
169         "b1": "Fixed tripod or mount.",
170         "b2": "<b>No panning, no zooming</b> during play."
171     },
172     "step4": {
173         "title": "Light & record",
174         "b1": "Even indoor lighting; avoid backlight, glare and very dark halls.",
175         "b2": "Record the <b>whole session</b> and upload it as-is ? KhelSutra does the
cutting."
176     },
177     "doTitle": "Do",
178     "do1": "Fixed camera, full court, behind the baseline",
179     "do2": "1080p+, landscape, audio on",
180     "do3": "One match on one court",
181     "dontTitle": "Avoid",
182     "dont1": "Handheld or phone-in-hand",
183     "dont2": "Broadcast-style pans & zooming",
184     "dont3": "Several courts in one frame; very dark halls",
185     "consent": "Please record adults with their okay. Footage involving minors is
never used to improve the tool and is never published.",
186     "questions": "Questions?",
187     "printTip": "Tip: press Ctrl/Cmd + P to print or save as PDF."
188 },
189 "ownModel": {
190     "requestAccess": "Request access ?"
191 }
192 },
193 "selection": {
194     "count": "{count, plural, one {# rally selected} other {# rallies selected}}"
195 }

```

196 }
197

8. user (2026-06-21T18:49:44.294Z)

```
1      {  
2        "app": {  
3          "badge": "????-???? ???? · ????",  
4          "title": "???? ????????? ????????????? ?????",  
5          "intro": "<b>????? ????</b>? ????????? ?????? ???? ? ???. <i>???? ????? ????  
15?</i> ? ??? ?????????????? ?????? ??? ?????????? ??????. ?????? <b>?????????</b>  
????????????????; ?????????????? ???? ??????????."   
6        },  
7        "lang": {  
8          "label": "????",  
9          "en": "English",  
10         "kn": "?????",  
11         "hi": "?????",  
12         "es": "Español",  
13         "da": "Dansk",  
14         "id": "Bahasa Indonesia"  
15       },  
16       "bulk": {  
17         "selectAll": "????????? ??????????",  
18         "clear": "????????????",  
19         "invert": "?????????"  
20       },  
21       "mode": {  
22         "demo": "?? ???? ????????? ? ?????? ?????????? ??? ?????? ??? ??????  
????????????????.",  
23         "loading": "???? ?????????? ?????????? ??????????????",  
24         "error": "? ?????? ?????????? ??? ??????????: {error}",  
25         "loaded": "???? ?????????? <b>{count}</b> ????????? ??????????"  
26       },  
27       "query": {  
28         "placeholder": "???. over 10s · top 15 longest · shortest first",  
29         "ariaLabel": "????, ??? ???? ???? ?????????? ?????????? ?????? ????",  
30         "apply": "?????????",  
31         "notAvailable": "???? ??????????",  
32         "notAvailableTitle": "{token}? ?? {kind} ????? ???? ? ????? ??????????",  
33         "notAvailableHelp": "???? ??????, ?????? ?????? ?????? ?????????? ? ??????  
????? ???? ?????????? ?????? ????",  
34         "unreadable": "????? ?????????? ? ?over 10s? ??? ?top 15 longest? ??????????",  
35         "hint": "10 ?????????? ???? ????????????? over 10s ??? ???? ? ???? ??????  
????????? ?????? ????",  
36         "try": "???? ??????????:",  
37         "support": "<b>????? ????</b>? ????????? ?????? ????????? ??????????  
?????????????. ???-???? <x>?????????</x> (????????????????)."  
38       },  
39       "grid": {  
40         "empty": "????? ???? ?????? ??????????????",  
41         "rally": "????? {n}",  
42         "inReel": "????? {n}, ??????????",  
43         "notInReel": "????? {n}, ??????????????",  
44         "start": "?????????",  
45         "long": "????",  
46         "shots": "~{count} ?????? · ??????????",  
47         "shotsTitle": "???? ???? ?????????????? (????????????????) ????  
????????????{conf}. ?????? ???? ???? ?????????? ?????? ????",  
48         "shotsConfidence": " ? ??????: {conf}"  
49       },  
50       "footer": {  
51         "count": "{count, plural, one {# ?????} other {# ?????????}}",  
52         "summaryRest": "????????? · ??? {total}",  
53         "reelName": "??? ?????",
```



```
54     "reelNamePlaceholder": "my-highlights",
55     "build": "???? ????? ({count})",
56     "building": "????????????????",
57     "floorOne": "<b>????? ???? ?????????? ??????</b>????? {min} ?????? ?????? ??????????
? ?????? ?????? ?????????????????, ?????????? ??? ?????????????????? ?????? ?????????????? ??????",
58     "floorAll": "<b>????????????? ?????? {count} ??????????????</b> {min} ?????? ?????
??????????? ? ?????? ?????????????? ?????????????????, ?????????? ??? ?????????????????? ?????
????????????? ??????",
59     "floorSomeLead": "{count, plural, one {# ??????} other {# ?????????}}",
60     "floorSomeRest": "{min} ?????? ?????? ?????????????? ?????????? ?????????????????????
(????????? <code>min_shot_count</code> ????)".
61 },
62     "result": {
63         "ready": "????? ??? ?????????????",
64         "failed": "???? ??????????",
65         "building": "????? ??? ??????????????????",
66         "buildingReassure": "???? ?????????? ?????????????? ??? ?????????? ?????????????????? ?
????????????? ?????? ?????????? ??????????".
67         "dismiss": "?????????",
68         "download": "{filename} ?????????? ????"
69     },
70     "build": {
71         "noVideo": "??? ?????? ?????????? ? ??? ?????? ?????? ?????? ?????? ??????????????????".
72     },
73     "ingest": {
74         "title": "?????? ??????????????????",
75         "intro": "????????????????? ??? ?????????? ?????? ? ?????? ?????? ??? ??????????????????
???? ? ??? ?????????????? ?????????????? ?????? ?????? ?????????? ??????????????",
76         "uriPlaceholder": "gs://your-bucket/match.mp4",
77         "uriAriaLabel": "?????? ??? ? gs:// ?????? URL ??? ?????? ?????????????????? ??",
78         "hint": "gs:// ?????? ?????, ??? ?????? ?????????????????? ?????? ?????? ??????
?????????",
79         "submit": "?????? ??????????????????",
80         "submitting": "????????????????????",
81         "cancel": "?????????????",
82         "working": "????? ?????? ??????????????????????????",
83         "progress": "{progress}",
84         "retry": "????? ??????????",
85         "reassure": "????????? ?????????? ? ?????? ?????? ?????????????????? ?????????? ?????????????
???? ??????????????????; ?????? ?????????? ?????? ?????????? ??????????????",
86         "statusFailed": "????????????????????????????",
87         "chooseFile": "?????? ??? ??????",
88         "uploadAria": "????????? ?????? ?????? ??? ??????",
89         "fileHint": "? ?????????? .mp4 ??? .mov ?????? ? ??? ?????????? ?????????, ???
????????????? ?????????????? ??????????????",
90         "orLink": "???? ?????? ?????? ????",
91         "uploadingStatus": "????????? ?????????????",
92         "uploadProgress": "????????? ??????"
93     },
94     "errors": {
95         "networkError": "?????? ??????????????????. ??? ??????????????????",
96         "invalidUri": "????? ?????? ?????????? ?????????? ? gs:// ?????? ??? ??????????????????
??.",
97         "emptyResult": "????????? ?????????? ??? ? ?????????????? ?????? ?????????? ??????????
????????? ?????? ?????????????? ??????????????",
98         "timeout": "????????????????? ?????? ??? ??????????????????????????. ??? ?????? ?????????????????????
? ?????? ?????????????????? ?????????? ???, ??? ?????? ??????????????",
99         "engineGeneric": "??? ??????????. ?????????? ?????? ??????????????"
100     },
101     "tap": {
102         "title": "????????? ?????? ?????? ????",
103         "longest": "??? ?????????? ??????",
104         "shortest": "??? ?????????? ??????",
105         "over10": "10 ?????????????? ??????",
106         "over20": "20 ?????????????? ??????"
```

```
107     "top5": "??? ????? 5",
108     "top10": "??? ????? 10"
109 },
110 "site": {
111     "nav": {
112         "how": "??? ??? ???? ?????????",
113         "why": "??? ?? ?????",
114         "ownModel": "?????? ?????",
115         "flyer": "???????",
116         "roadmap": "???????",
117         "faq": "????????????",
118         "requestAccess": "?????? ??"
119     },
120     "hero": {
121         "eyebrow": "?????-???? ???? . ??????",
122         "lead": "KheISutra ???? ????????????? ?????????? ?????, ???-?????,
????? ?????? ????????????? ?????????????? ? ?????? ?????? ?????????? ??? ?????? ?????????? ??????????
????? ?????? ??? ?????????? ?????? ?????????? ????, ?????????????? ????",
123         "ctaPrimary": "????? ?????? ????",
124         "ctaHow": "??? ??? ???? ?????????? ??"
125     },
126     "record": {
127         "kicker": "?????? ?????????? ????",
128         "title": "?? ???? ???? ? ?????????? ?????????? ??????????",
129         "intro": "????? ?????? ?????????????????? ??????????????. ?????????????? ???
????? ?????????? ??? ? ?????????????? ?????? KheISutra ?? ??? ???? ??????????",
130         "goldenRuleLead": "????? ????: ??? ?????, ??? ?????, ??? ?????????????? ?
????? ?????????????? ??????????????",
131         "goldenRuleRest": "??? ?????? ?????????????? ?????? ??? ?????????? ??????
?????????; ??? ?????????? ?????? ?????? ??? ?????? ?????????????????? ??? ??????",
132         "place": {
133             "title": "????????? ?????",
134             "body": "<b>??? ?????????</b> ????, ?????????? (????? ?????????? ??? ??????),
~30° ?????? ???? ? <b>??? ?????</b> ?????????????, ?????? ?????? ??????????????????."
135         },
136         "settings": {
137             "title": "?????????????",
138             "body": "<b>1080p+</b>, ??????????????, ?????????????? <b>30 fps</b> (?????
????????? 60 fps). <b>Audio ON</b> ? ?????? ?????? ??? ?????????????? ?????? ??????????????."
139         },
140         "still": {
141             "title": "????????? ?????",
142             "body": "????? ?????????? ??? ?????. ?? ?????? <b>????? ???, ??? ????</b> ?
????????, ???-????? ?????????? ? ?????????? ??????????????"
143         },
144         "upload": {
145             "title": "????????? ??? ???? ?????????? ????",
146             "body": "????? ????, ??? <b>??? ?????</b> ?????? ?????????? ??? ??????????
????????? ??? ? KheISutra ?????????????? ?????? ?????????? ??? ??????????"
147         },
148         "checklistCta": "???????? 1-???? ?????????? ?????????? ??"
149     },
150     "capture": {
151         "title": "1-???? ?????????? ?????????? ?????",
152         "intro": "????? ?????????? ?????????, ???-????? ?????????????????? ??????????????
??? ?????? ?????? ???? ?????????????? ?????? ?????????? ??????????",
153         "ruleLead": "????? ????: ??? ?????, ??? ?????, ??? ?????????????? ? ?????
????????????????? ??????????????",
154         "ruleRest": "??? ?????????? ?????? ??? ?????? ?????????????????? ??? ?????; ???
????? ?????????????? ?????? ??? ?????????? ?????????? ??????????",
155         "step1": {
156             "title": "????????? ?????",
157             "b1": "<b>??? ?????????</b> ????, ?????????? ? ?????? ?????????, ??? ????
????????????, ??? ??????",
158             "b2": "????? <b>30°</b> ?????? ?????; ?????? ??? ?????? ??? ??????"
```


camera</h3>

87: <li data-i18n-html="site.capture.step1.b1">Behind one baseline, elevated ? a high tripod, a few metres up, or a balcony.

88: <li data-i18n-html="site.capture.step1.b2">Tilt down about 30°; keep the whole court in frame with a little margin and net headroom.

89: <li data-i18n="site.capture.step1.b3">Players should never leave the frame.

93: <h3>2</h3>Settings</h3>

95: <li data-i18n-html="site.capture.step2.b1">1080p or higher, shot in landscape.

96: <li data-i18n-html="site.capture.step2.b2">30 fps for badminton/pickleball; 60 fps for table tennis.

97: <li data-i18n-html="site.capture.step2.b3">Audio ON (don't mute) ? keep the mic unobstructed, wind/noise reduction off.

101: <h3>3</h3>Keep it still</h3>

103: <li data-i18n="site.capture.step3.b1">Fixed tripod or mount.

104: <li data-i18n-html="site.capture.step3.b2">No panning, no zooming during play.

108: <h3>4</h3>Light & record</h3>

110: <li data-i18n="site.capture.step4.b1">Even indoor lighting; avoid backlight, glare and very dark halls.

111: <li data-i18n-html="site.capture.step4.b2">Record the whole session and upload it as-is ? KhelSutra does the cutting.

118: <h4 data-i18n="site.capture.doTitle">Do</h4>

120: <li data-i18n="site.capture.do1">Fixed camera, full court, behind the baseline

121: <li data-i18n="site.capture.do2">1080p+, landscape, audio on

122: <li data-i18n="site.capture.do3">One match on one court

126: <h4 data-i18n="site.capture.dontTitle">Avoid</h4>

128: <li data-i18n="site.capture.dont1">Handheld or phone-in-hand

129: <li data-i18n="site.capture.dont2">Broadcast-style pans & zooming

130: <li data-i18n="site.capture.dont3">Several courts in one frame; very dark halls

135: <p class="consent" data-i18n="site.capture.consent">Please record adults with their okay. Footage involving minors is never used to improve the tool and is never published.</p>

138: Questions? hello@khelsutra.guru . khelsutra.guru

139: Tip: press Ctrl/Cmd + P to print or save as PDF.

11. user (2026-06-21T18:49:52.221Z)

23:<link href="https://fonts.googleapis.com/css2?family=Plus+Jakarta+Sans:wght@400;500;600;700;800&family=Noto+Sans+Kannada:wght@400;500;600;700&family=Noto+Sans+Devanagari:wght@400;500;600;700&display=swap" rel="stylesheet" />

31: --font:'Plus Jakarta Sans','Noto Sans Kannada','Noto Sans Devanagari',system-ui,-apple-system,Segoe UI,Roboto,Helvetica,Arial,sans-serif;

356: <p data-i18n-html="site.record.place.body">Behind one baseline, elevated (a high tripod or a balcony), tilted down ~30° ? the whole court in frame, players never cropped.</p>

361: <p data-i18n-html="site.record.settings.body">1080p, landscape, 30 fps for badminton (60 fps for table tennis). Audio ON ? the shuttle's ?thwack? helps find rallies.</p>

366: <p data-i18n-html="site.record.still.body">Fixed tripod or mount. No panning, no zooming during play ? a steady, full-court view is what the tool is tuned for.</p>

371: <p data-i18n-html="site.record.upload.body">Even lighting, then record the whole session and upload it as-is ? KhelSutra finds the rallies and does the cutting.</p>

12. assistant (2026-06-21T18:49:56.994Z)

I now have all the context I need. Let me analyze test coverage gaps for A3 specifically.

Let me verify a few specific concerns. First, the drift test for `data-i18n-html` keys ? the assets test (line 261-262) checks `enCatalog` has `**` markup but never checks the catalog value matches the HTML default for the rich keys. Let me confirm there's a real drift risk by comparing one.**

13. user (2026-06-21T18:49:58.450Z)

```
350         </div>
351
352         <div class="grid g4">
353             <div class="card">
354                 <div class="ic" aria-hidden="true">?</div>
355                 <h3 data-i18n="site.record.place.title">Place the camera</h3>
356                 <p data-i18n-html="site.record.place.body">Behind <b>one baseline</b>, elevated
(a high tripod or a balcony), tilted down ~30° ? the <b>whole court</b> in frame, players never
cropped.</p>
357             </div>
358             <div class="card">
359                 <div class="ic" aria-hidden="true">?</div>
360                 <h3 data-i18n="site.record.settings.title">Settings</h3>
361                 <p data-i18n-html="site.record.settings.body"><b>1080p</b>, landscape, <b>30
fps</b> for badminton (60 fps for table tennis). <b>Audio ON</b> ? the shuttle's ?thwack? helps
find rallies.</p>
362             </div>
363             <div class="card">
364                 <div class="ic" aria-hidden="true">?</div>
365                 <h3 data-i18n="site.record.still.title">Keep it still</h3>
366                 <p data-i18n-html="site.record.still.body">Fixed tripod or mount. <b>No panning,
no zooming</b> during play ? a steady, full-court view is what the tool is tuned for.</p>
367             </div>
368             <div class="card">
369                 <div class="ic" aria-hidden="true">?</div>
370                 <h3 data-i18n="site.record.upload.title">Record & upload</h3>
371                 <p data-i18n-html="site.record.upload.body">Even lighting, then record the
<b>whole session</b> and upload it as-is ? KhelSutra finds the rallies and does the cutting.</p>
372             </div>
373         </div>
374
```

14. assistant (2026-06-21T18:50:15.843Z)

I now have a complete picture. Let me synthesize the test-coverage findings.

Key observations:

1. ****`data-i18n-html` fallback-on-missing-key IS covered**** (shim test line 84-85, `does.not.exist`). Don't re-propose.
2. ****kn render of a rich `**` key end-to-end IS covered**** (shim test line 87-89). Don't re-propose.**
3. ****Drift between HTML default and en catalog for the rich `data-i18n-html` keys is NOT covered.**** The existing drift test (assets test line 224-231) only checks two **plain** `data-i18n` keys (`site.capture.title`, `site.nav.requestAccess`). The A3 test at line 261-262 only asserts the en catalog **contains** `**` markup ? it never compares the en catalog value against the HTML default innerHTML for the rich keys. So if `index.html`'s `site.record.settings.body` default text diverges from `enCatalog.site.record.settings.body`, an en visitor would silently see a different (overwritten) string than the no-JS default. This is a real gap, and the rich keys are the most fragile (the `**` markup must match byte-for-byte including curly quotes/em-dashes).****
4. ****`data-i18n-attr` + missing key fallback combination is covered**** for the kn value case but the en-fallback path through `data-i18n-attr` is fine.
5. ****Font-stack robustness****: assets test line 239 asserts `--font:...`Noto Sans

Kannada'...'Noto Sans Devanagari'` but it does NOT assert order relative to a Latin font, nor that Kannada precedes Devanagari, nor (more importantly) that `display=swap` / the actual `` weights are present ? though line 237-238 covers the family load. The regex `--font:[^;]\*` would also pass if Noto came *after* `system-ui`, which would defeat Indic glyph selection on some systems. Minor robustness gap.

6. ****Plurals in the kn/hi site.* recording namespace****: none exist, so N/A.

7. ****`interpolate`/`translate` on a rich html value with interpolation****: the rich keys have no `{vars}`, so N/A.

8. ****The `data-i18n-html` + `data-i18n-attr` interaction****: in the shim, if both `data-i18n-html` and `data-i18n-attr` are present, `innerHTML` wins (the `if (htmlKey)` branch precedes the `attr` branch). Not tested, but no A3 element combines them, so low value.

Let me confirm the drift gap is real and not covered elsewhere by checking if any test compares a rich key's HTML default to its catalog value.

15. user (2026-06-21T18:50:16.495Z)

```
261: assert.match(enCatalog.site.record.place.body, /<b>.*<\b>/s, "en record.place.body lost its <b> markup");
```

```
262: assert.match(enCatalog.site.capture.step2.b1, /<b>1080p or higher<\b>/, "en capture.step2.b1 markup drift");
```

16. assistant (2026-06-21T18:50:20.895Z)

Confirmed. The drift between the HTML `data-i18n-html` default innerHTML and the en catalog value is not asserted anywhere. The existing checks only verify the catalog *contains* markup, not that it *equals* the HTML default. This is the most material gap. Now I'll report the findings.

17. user (2026-06-21T18:50:41.885Z)

Structured output provided successfully

18. assistant (2026-06-21T18:50:56.343Z)

I reviewed the two test files against the A3 surface (shim, both HTML pages, en/kn catalogs). Coverage is solid on the obvious cases ? `data-i18n-html` fallback-on-missing-key, kn end-to-end render of a rich `**` key, and the anti-seed/in-script checks are all present, so I did not re-propose those.**

Three real gaps, reported via StructuredOutput:

1. ****Major ? rich-key drift is unguarded.**** The "no drift between HTML default and en catalog" test only covers two *plain* `data-i18n` keys. The A3 test only asserts the en catalog *contains* `**` markup; nothing asserts the HTML `data-i18n-html` default innerHTML *equals* the en catalog value. The rich keys (curly quotes, em-dashes, multiple `**` tags) are the most drift-prone, and divergence would silently overwrite the no-JS default for en visitors with CI green. Fix: JSDOM-walk every `[data-i18n-html]` and assert `el.innerHTML.trim() === enCatalog[key]`.****

2. ****Minor ? font-stack assertion ignores order.**** `--font:... 'Noto...'` would still pass if Noto were placed after `system-ui`, defeating Indic glyph selection (the tofu case A3 exists to prevent). Fix: pin order so Noto families precede `system-ui`.

3. ****Minor ? the `data-i18n-html` en-fallback rung is untested.**** Tests cover en-direct, kn-direct, and total-miss, but not "kn lacks a rich key, en has it" ? should set innerHTML from en with `**` intact. Fix: a synthetic-catalog `applyDom` case.**

Relevant files: `C:/Users/avidu/Projects/khelsutra-fu-wt/site/src/assets.test.js`,
`C:/Users/avidu/Projects/khelsutra-fu-wt/site/src/i18n.shim.test.js`.